

1. Prob 16.8

(a) The covariant equation of motion from problem 16.7 reads

$$\dot{p}^\mu = F_{\text{ext}}^\mu + \frac{2e^2}{3mc^3} \left[\ddot{p}^\mu + \frac{p^\mu (\dot{p}_\nu \dot{p}^\nu)}{m^2 c^2} \right] \quad (1)$$

Note here F_{ext}^μ is the "covariant force" in the sense it is the derivative of the 4-momentum with respect to the proper time (to be accurate, this is the part of the momentum change contributed from the external force, not the entire $dp^\mu/d\tau$). But in this problem $f(\tau)$ represents the ordinary force that is the derivative of the 1D momentum with respect to the lab time. Since $dt = \gamma d\tau$, we have

$$F_{\text{ext}}^1 = \gamma f(\tau) = \sqrt{1 + \frac{p^2}{m^2 c^2}} f(\tau) \quad (2)$$

For the 1D problem, we take $\mu = 1$, the contraction $\dot{p}_\nu \dot{p}^\nu$ can be calculated via

$$\dot{p}^\nu = \frac{dp^\nu}{d\tau} = \frac{d}{d\tau} (\sqrt{p^2 + m^2 c^2}, p, 0, 0) = \left(\frac{p\dot{p}}{\sqrt{p^2 + m^2 c^2}}, \dot{p}, 0, 0 \right) \quad (3)$$

so

$$\dot{p}_\nu \dot{p}^\nu = \left(\frac{p\dot{p}}{\sqrt{p^2 + m^2 c^2}} \right)^2 - \dot{p}^2 = - \left(\frac{m^2 c^2 \dot{p}^2}{p^2 + m^2 c^2} \right) \quad (4)$$

Putting (4) and (2) into (1) gives the desired equation of motion

$$\dot{p} - \frac{2e^2}{3mc^3} \left(\ddot{p} - \frac{p\dot{p}^2}{p^2 + m^2 c^2} \right) = \sqrt{1 + \frac{p^2}{m^2 c^2}} f(\tau) \quad (5)$$

(b) For $p = mc \sinh y$,

$$\dot{p} = mc \cosh y \dot{y} \quad \ddot{p} = mc (\sinh y \dot{y}^2 + \cosh y \ddot{y}) \quad (6)$$

and

$$\frac{p\dot{p}^2}{p^2 + m^2 c^2} = \frac{mc \sinh y (mc \cosh y \dot{y})^2}{m^2 c^2 \cosh^2 y} = mc \sinh y \dot{y}^2 \quad (7)$$

so equation (5) becomes

$$mc \cosh y \dot{y} - \frac{2e^2}{3mc^3} \cdot mc \cosh y \ddot{y} = \cosh y f(\tau) \quad \implies \quad mc (\dot{y} - \tau_0 \ddot{y}) = f(\tau) \quad (8)$$

This is the Abraham-Lorentz equation (16.9) with v replaced by cy .

The differential equation (8) can be rewritten as

$$\frac{\dot{y}}{\tau_0} - \ddot{y} = \frac{f(\tau)}{mc \tau_0} \quad \implies \quad \frac{d}{d\tau} (e^{-\tau/\tau_0} \dot{y}) = -\frac{e^{-\tau/\tau_0} f(\tau)}{mc \tau_0} \quad (9)$$

giving the general solution

$$\dot{y}(\tau) = e^{\tau/\tau_0} \left[\dot{y}(0) - \frac{1}{mc \tau_0} \int_0^\tau e^{-s/\tau_0} f(s) ds \right] \quad (10)$$

which diverges as $\tau \rightarrow \infty$ unless we enforce

$$\dot{y}(0) = \frac{1}{mc \tau_0} \int_0^\infty e^{-s/\tau_0} f(s) ds \quad (11)$$

With this, we can write

$$\dot{y}(\tau) = \frac{e^{\tau/\tau_0}}{mc \tau_0} \int_\tau^\infty e^{-s/\tau_0} f(s) ds = \frac{1}{mc} \int_0^\infty e^{-\xi} f(\tau + \tau_0 \xi) d\xi \quad (12)$$

Thus

$$y(\tau) = y_0 + \int_0^\tau \dot{y}(s') ds' \quad \text{where } y_0 = \sinh^{-1}\left(\frac{p_0}{mc}\right) \quad (13)$$

In summary, the general solution is

$$p(\tau) = mc \sinh y(\tau) \quad \text{where} \quad y(\tau) = \sinh^{-1}\left(\frac{p_0}{mc}\right) + \frac{1}{mc} \int_0^\tau \int_0^\infty e^{-\xi} f(s' + \tau_0 \xi) d\xi ds' \quad (14)$$

2. Prob 16.9

(a) Since $p^\nu p_\nu$ is a conserved quantity, we have the orthogonality condition

$$p^\nu \dot{p}_\nu = 0 \quad (15)$$

Taking the derivative again gives

$$\dot{p}^\nu \dot{p}_\nu = -\ddot{p}^\nu p_\nu \quad (16)$$

Replacing this into the F_μ^{rad} expression gives

$$\begin{aligned} F_\mu^{\text{rad}} &= \frac{2e^2}{3mc^3} \left[\frac{d^2 p_\mu}{d\tau^2} + \frac{p_\mu}{m^2 c^2} \left(\frac{dp_\nu}{d\tau} \frac{dp^\nu}{d\tau} \right) \right] \\ &= \frac{2e^2}{3mc^3} \left(\frac{d^2 p_\mu}{d\tau^2} - \frac{p_\mu p_\nu}{m^2 c^2} \frac{d^2 p^\nu}{d\tau^2} \right) \\ &= \frac{2e^2}{3mc^3} \left[\left(g_{\mu\nu} - \frac{p_\mu p_\nu}{m^2 c^2} \right) \frac{d^2 p^\nu}{d\tau^2} \right] \end{aligned} \quad (17)$$

(b) With the replacement $d^2 p^\nu / d\tau^2 \rightarrow g^{\nu\lambda} dF_\lambda^{\text{ext}} / d\tau$, (17) becomes

$$F_\mu^{\text{rad}} = \frac{2e^2}{3mc^3} \left[\left(g_{\mu\nu} - \frac{p_\mu p_\nu}{m^2 c^2} \right) g^{\nu\lambda} \frac{dF_\lambda^{\text{ext}}}{d\tau} \right] = \tau_0 \left(\frac{dF_\mu^{\text{ext}}}{d\tau} - \frac{p_\mu p_\nu}{m^2 c^2} \frac{dF_\nu^{\text{ext}}}{d\tau} \right) \quad (18)$$

The equation of motion from problem 16.7 now reads

$$\frac{dp^\mu}{d\tau} = F_\mu^{\text{ext}} + \tau_0 \left(\frac{dF_\mu^{\text{ext}}}{d\tau} - \frac{p^\mu p_\nu}{m^2 c^2} \frac{dF_\nu^{\text{ext}}}{d\tau} \right) \quad (19)$$

With $dt = \gamma d\tau$, using the same argument as in problem 16.8 (a), we can write the covariant force in terms of the ordinary external force vector $\mathbf{F}$ as

$$F_{\text{ext}}^\nu = \gamma \left(\frac{\mathbf{F} \cdot \mathbf{v}}{c}, \mathbf{F} \right) \quad (20)$$

Let us first calculate the contraction $p_\nu dF_{\text{ext}}^\nu / d\tau$ in (19). Notice that

$$p_\nu F_{\text{ext}}^\nu = \gamma (mc, m\mathbf{v}) \cdot \gamma (\mathbf{F} \cdot \mathbf{v}/c, \mathbf{F}) = 0 \quad (21)$$

Taking the derivative $d/d\tau$ of the above gives

$$p_\nu \frac{dF_{\text{ext}}^\nu}{d\tau} = -\frac{dp_\nu}{d\tau} F_{\text{ext}}^\nu = -\gamma \left(\frac{d(\gamma mc)}{dt}, \frac{d(\gamma m\mathbf{v})}{dt} \right) \cdot \gamma \left(\frac{\mathbf{F} \cdot \mathbf{v}}{c}, \mathbf{F} \right) \quad (22)$$

Using the fact that

$$\frac{d\gamma}{dt} = \frac{d}{dt} (1 - \boldsymbol{\beta} \cdot \boldsymbol{\beta})^{-1/2} = \gamma^3 \boldsymbol{\beta} \cdot \frac{d\boldsymbol{\beta}}{dt} \quad (23)$$

(22) becomes

$$p_\nu \frac{dF_{\text{ext}}^\nu}{d\tau} = -m\gamma^2 \left(\gamma^3 c \boldsymbol{\beta} \cdot \frac{d\boldsymbol{\beta}}{dt}, \gamma^3 \left(\boldsymbol{\beta} \cdot \frac{d\boldsymbol{\beta}}{dt} \right) \mathbf{v} + \gamma \frac{d\mathbf{v}}{dt} \right) \cdot \left(\frac{\mathbf{F} \cdot \mathbf{v}}{c}, \mathbf{F} \right) = m\gamma^3 \left(\mathbf{F} \cdot \frac{d\mathbf{v}}{dt} \right) \quad (24)$$

Putting (24) into (19) and taking the spatial part, we have

$$\gamma \frac{d\mathbf{p}}{dt} = \gamma \mathbf{F} + \tau_0 \left[\gamma \frac{d(\gamma \mathbf{F})}{dt} - \frac{\mathbf{p}}{m^2 c^2} \cdot m\gamma^3 \left(\mathbf{F} \cdot \frac{d\mathbf{v}}{dt} \right) \right] \quad (25)$$

which reduces to

$$\frac{d\mathbf{p}}{dt} = \mathbf{F} + \tau_0 \left[\gamma^3 \left(\boldsymbol{\beta} \cdot \frac{d\boldsymbol{\beta}}{dt} \right) \mathbf{F} + \gamma \frac{d\mathbf{F}}{dt} - \gamma^3 \boldsymbol{\beta} \left(\mathbf{F} \cdot \frac{d\boldsymbol{\beta}}{dt} \right) \right] = \mathbf{F} + \tau_0 \left[\gamma \frac{d\mathbf{F}}{dt} - \frac{\gamma^3}{c^2} \frac{d\mathbf{v}}{dt} \times (\mathbf{v} \times \mathbf{F}) \right] \quad (26)$$

3. Prob 16.10

- (a) (16.9) as a differential equation for $\mathbf{v}$ is the same differential equation (8) for y as problem 16.8 (b), so all the arguments there apply here. In particular, no-runaway condition enforces

$$m\dot{\mathbf{v}}(t) = \int_0^\infty e^{-s} \mathbf{F}(t + \tau s) ds \quad (27)$$

- (b) Expanding $\mathbf{F}(t + \tau s)$ around t gives

$$m\dot{\mathbf{v}}(t) = \int_0^\infty e^{-s} \sum_{n=0}^\infty \frac{\tau^n s^n}{n!} \frac{d^n \mathbf{F}(t)}{dt^n} ds = \sum_{n=0}^\infty \frac{\tau^n}{n!} \frac{d^n \mathbf{F}(t)}{dt^n} \underbrace{\int_0^\infty e^{-s} s^n ds}_{n!} = \sum_{n=0}^\infty \tau^n \frac{d^n \mathbf{F}(t)}{dt^n} \quad (28)$$

- (c) For the step force $F(t) = F_0 \Theta(t)$ in one dimension, part (a) gives

$$ma(t) = F_0 \int_0^\infty e^{-s} \Theta(t + \tau s) ds = \begin{cases} F_0 e^{t/\tau} & t < 0 \\ F_0 & t \geq 0 \end{cases} \quad (29)$$

The product of mass and velocity is given by

$$mv(t) = \int_{-\infty}^t ma(t') dt' = \begin{cases} F_0 \tau e^{t/\tau} & t < 0 \\ F_0 \tau (1 + t/\tau) & t \geq 0 \end{cases} \quad (30)$$

Both a and v are continuous at the origin. The particle starts to move before the force is switched on: the acceleration builds up as $e^{t/\tau}$ over a window $\sim \tau$ ahead of $t = 0$ (pre-acceleration). This is the price of the no-runaway condition imposed in part (a): as in problem 16.8 (b), eliminating the runaway of the Abraham–Lorentz equation (16.9) by a future boundary condition (e.g., $t = \infty$) is exactly what makes the acceleration depend on the force at later times, producing the acausal precursor.

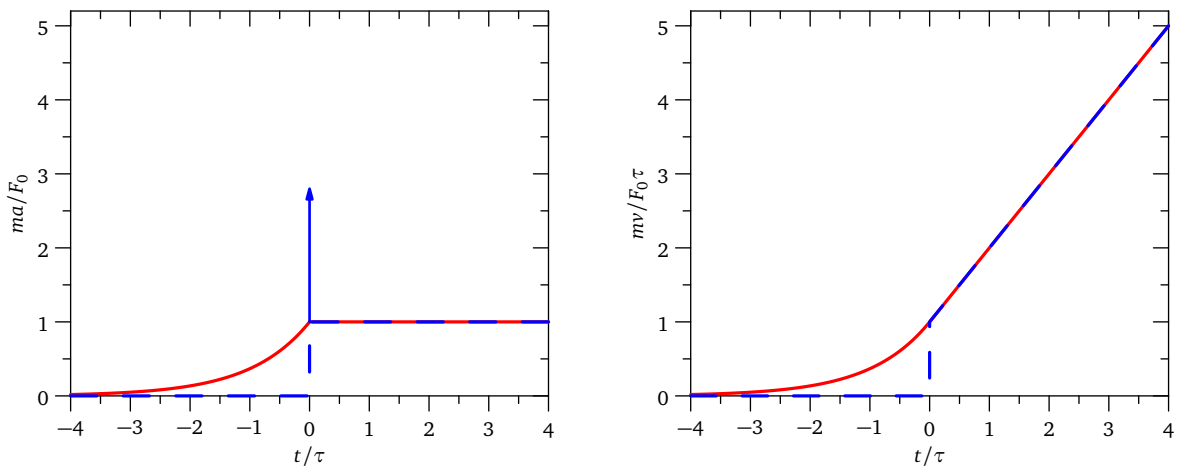
By contrast, for the truncated equation (16.10),

$$m\dot{\mathbf{v}} = \mathbf{F} + \tau \dot{\mathbf{F}} \quad (31)$$

has $\dot{\mathbf{F}} = F_0 \delta(t)$, so

$$ma(t) = F_0 \Theta(t) + \tau F_0 \delta(t) \quad \Rightarrow \quad mv(t) = F_0 (t + \tau) \Theta(t) \quad (32)$$

This reproduces the exact answer for $t > 0$, but replaces the smooth exponential precursor by zero pre-acceleration plus an unphysical δ -function impulse at $t = 0$, which makes the velocity jump discontinuously from 0 to $F_0 \tau$. The full integral equation removes both the runaway and the impulse, spreading the response smoothly over the tiny time $\tau \sim 10^{-23}$ s.



— exact (integral eq.) — approx (16.10)